

CoverMap

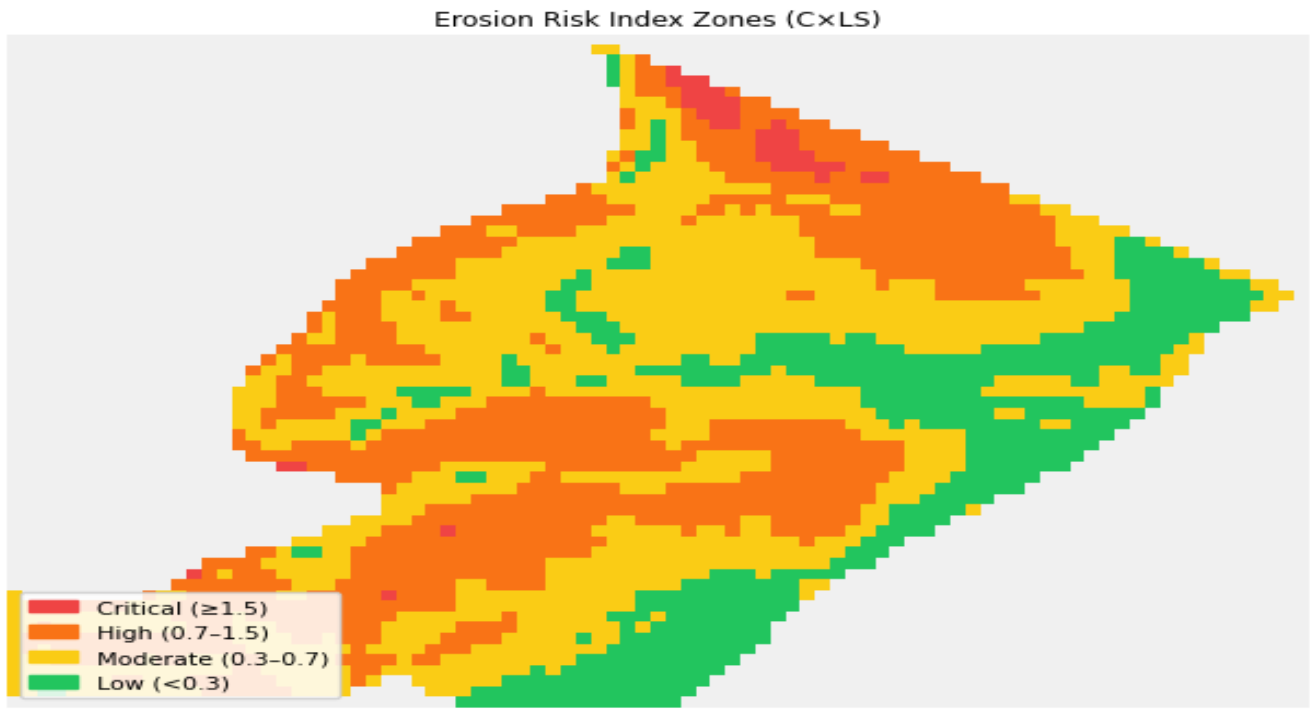
Stephen Zimmerman, CCA MS
Ankeny, IA | Ag Research Scientist

Field Cover Crop Assessment

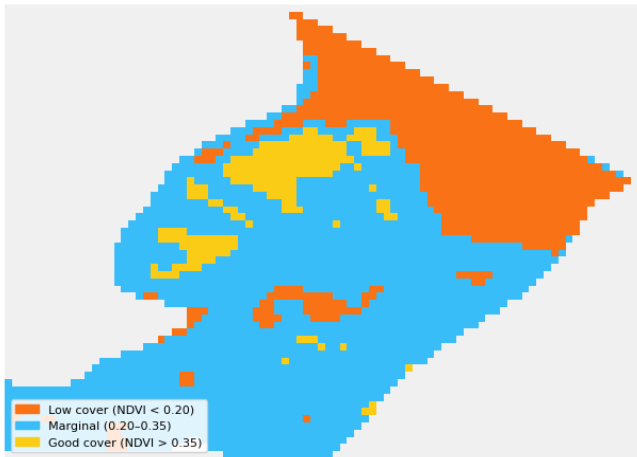
Field: Bin Field (2)	Farm: Bizy Farms Inc.	County: Shelby County, IA	Report Date: May 15, 2026
Previous crop: Corn	Termination date: May 1st.	Dominant soil series: Monona — K-factor: .32	Previous crop / tillage: Tillage — > 30% residue (conservation tillage)

NDVI: Apr 05, 2026 – Apr 20, 2026 | DEM: Iowa 3-meter Digital Elevation Model (Iowa DNR)

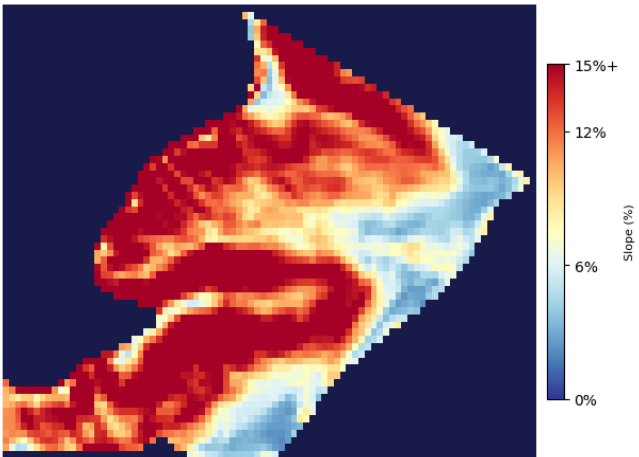
Field Risk Maps



Erosion Risk Index Zones (C×LS) — pixel-level RUSLE risk classification



NDVI Cover Quality — Low (< 0.20) / Marginal / Good (> 0.35)



Terrain Slope (% gradient) — Flat / Moderate / Steep

CoverMap Advisory & Recommendation

Erosion Concern: Moderate

Cover crop stand is variable across slope positions based on NDVI analysis. Erosion risk increases on steeper units during spring rainfall events.

Cover Crop Stand — NDVI Zone Summary

Zone	Acres	% of Field
Low Cover (NDVI < 0.20)	12.9	27%
Marginal (NDVI 0.20–0.35)	31.9	66%
Good Cover (NDVI > 0.35)	3.6	7%
Total	48.3	100%

Erosion Risk Zone Summary (C×LS)

Zone	C×LS Range	Acres	% of Field
Critical Risk	> 1.5	0.7	2%
High Risk	0.7–1.5	17.2	36%
Moderate Risk	0.3–0.7	21.0	43%
Low Risk	< 0.3	9.4	19%

Risk Index = C-factor (NDVI) × LS-factor (slope). Boundary-masked field pixels only. Critical >1.5 · High 0.7–1.5 · Moderate 0.3–0.7 · Low <0.3

NDVI imagery dated 6 scenes composited: Apr 08, 2026 – Apr 18, 2026. Field conditions may have changed since image capture. This report documents satellite-observed conditions only.

Cover Crop Stand Assessment — Satellite Documentation

Requirement	Data Source	Status
Cover crop present	Sentinel-2 NDVI > 0.20	■ NDVI 0.251 — cover crop confirmed
Image date	GEE metadata	6 scenes composited: Apr 08, 2026 – Apr 18, 2026
Estimated biomass	NDVI proxy	~700–1650 lb/acre (±40% NDVI proxy)
30% ground cover	NDVI threshold	■ Estimated adequate cover zones based on NDVI threshold — field verification recommended

Satellite-verified cover crop status. Field verification recommended before termination.

Field Level Results

Metric	Value
NDVI Mean	0.251
NDVI Range	0.146 – 0.576
Slope Mean (%)	12.0%
C-Factor (exp. model)	0.185 (26% reduction vs. baseline)
Risk Index (C×LS)	0.592

Erosion Concern	Moderate
-----------------	----------

Metric	Value
Est. Cover Crop Erosion Reduction	23.4%
Baseline Soil Loss (no cover)	46.0 t/ac/yr
Est. Soil Saved (area-weighted)	10.8 t/ac/yr

Estimates based on RUSLE C-factor methodology. Residue multipliers are not yet RUSLE2-validated. ±10 pt uncertainty on reduction percentage.

NDVI Source: Sentinel-2 via Google Earth Engine (NDVI: Apr 05, 2026 – Apr 20, 2026) | DEM: Iowa 3-meter Digital Elevation Model (Iowa DNR) | Slope: computed in UTM meters (EPSG:26915)
C-Factor methodology: Iowa RUSLE lookup table — Laflen & Roose (1998), ISU Extension PM-1209. This report is advisory only and does not constitute an official NRCS determination.
CoverMap Field Report · Stephen Zimmerman, CCA MS · Sentinel-2 via Google Earth Engine · Iowa RUSLE C-factor calibration · May 15, 2026